

Unit 10 Lesson 3: Radian Measure



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define radian.
- ☐ I can explain the relationship between the length of the arc intercepted by an angle to the radius.



10.3.1 Warm-Up: Would You Rather?

Would you Rather...

Deposit \$5 in the bank and have it triple each week for 4 weeks?

OR

Deposit \$3 in the bank and have it quadruple each week for 4 weeks?

1. Which would you rather choose? Justify your reasoning using mathematics.



10.3.2 Exploration: Discovering Radian Measure

Angles that are measured in degrees can also be measured in **radians**. Measuring in radians is similar to measuring distance using other units, such as inches or feet. Radians can be written as a decimal or in pi (π) form.

With your partner, work through the Desmos activity “Discover Radian Measure.” Complete the steps to answer each question.

Take a moment to look over the activity and the information provided.

- Slider “ r ” adjusts the radius of the circle. The radii for the activity range from 0 to 3, increasing by 0.5.
- Slider “ c ” adjusts the central angle.
- The central angle, $C_{\text{centralAngle}}$, is measured in radians, written as a decimal. Radians for the activity increase by 0.25.

Drag slider r to the right and left to observe how the measurements of radius, central angle, and arc length change as the r value changes. These values are in rows 2, 5, and 6 respectively of the activity.

Hit play on slider c to watch how the measurements of radius, central angle, and arc length changes. These values are in rows 2, 5, and 6 respectively.

1. Adjust slider c , $C_{\text{centralAngle}}$, so that the arc length, A_{rclength} , equals the radius, r . Complete the table for three different measures of r .

Radius (r)			
$C_{\text{centralAngle}}(c)$			
A_{rclength}			

2. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

3. Adjust slider r , radius, so that the arc length, A_{rclength} , is twice the radius, r . Complete the table for three different measures of r .

Radius (r)			
$C_{\text{centralAngle}}(c)$			
A_{rclength}			

4. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

5. Complete the statements based on your work in questions 1-4.

A radian is a unit used to express the measure of an

angle. Radians are based on the number of ☐ angles
☐ radii

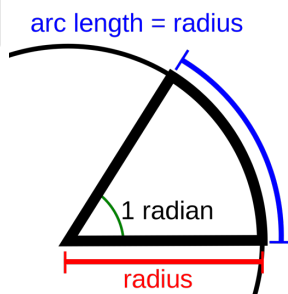
of a circle being wrapped along the ☐ diameter
☐ circumference

of the circle. The radian measure of a central angle is

equal to the length of the ☐ arc
☐ radius

intercepted by the central angle divided by the

- ☐ radius.
☐ arc length.

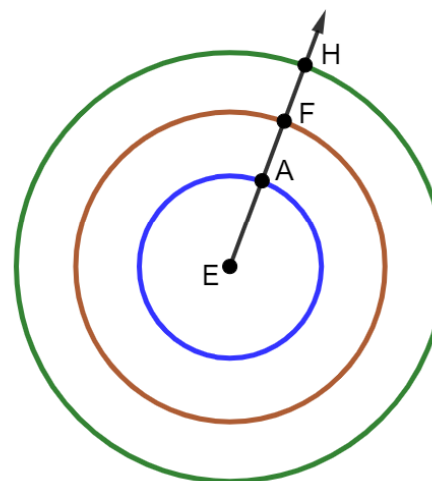


6. Describe how the diagram shows the relationship between the radius and arc length.



10.3.3 Exploration: Discovering Radians in a Whole Circle

1. Consider the diagram depicting three concentric circles each with center E . Point A lies on the blue Circle E , and has radius $EA = 1$ unit. Point F lies on the brown Circle E , and has radius $EF = 2$ units. Point H lies on the green Circle E , and has radius $EH = 3$ units.



- a. Complete the table with your partner.

Circle	Circumference	Ratio of Circumference to Radius
Radius $EA = 1$ unit		
Radius $EF = 2$ units		
Radius $EH = 3$ units		

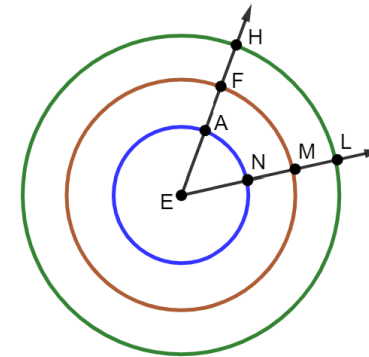
- b. Discuss any observations you made about the values in the table with your partner. Summarize your discussion.

- c. Write a conjecture about the ratio of the circumference to radius and a whole circle measured in radians.



10.3.4 Bring It Together: Relating Arc Length to the Radius

1. Concentric circles with center E are shown. Points A and N lay on the blue Circle E , and has radius $EA = 1$ unit. Points F and M lay on the brown Circle E , and has radius $EF = 2$ units. Points H and L lay on the green Circle E , and has radius $EH = 3$ units.



Describe how the arclengths AN , FM and HL are related.

2. Use words from the bank below to complete the statements.

angle arc radian radius

The length of the _____ intercepted by a(n) _____ is proportional to the _____ of the circle. The _____ measure of the angle is the constant of proportionality.



$$\text{Radians} = \frac{\text{Circumference}}{\text{Radius}} = \frac{2\pi r}{r} = \frac{2\pi}{1}$$

$$360^\circ = 2\pi \text{ radians}$$

$$180^\circ = \pi \text{ radians}$$



10.3.5 Wrap-Up: Cool Down

1. Your friend was absent today and will need notes from class. Write a quick note to help them understand relationship between the length of an arc intercepted by an angle and the radius.